TECHNICAL THOUGHT LEADERSHIP & MEDIUM PUBLICATIONS

Compiled by: Bassey Riman

EXECUTIVE SUMMARY: CONTRIBUTING TO THE DIGITAL TECHNOLOGY ECOSYSTEM

Beyond commercial and enterprise employment, Bassey Riman actively advances the wider digital technology sector through high-impact technical publications, educational tutorials, and open-source contributions. Through widely read articles published on Medium (@basseyriman), Bassey deconstructs non-trivial software engineering challenges?including progressive web application (PWA) offline architecture, dynamic mathematical UI scaling, deterministic hashing for UI components, and Next.js 15 animation orchestration?providing reusable code, architectural diagrams, and open-source implementation guides for software engineers and machine learning practitioners worldwide.

PUBLIC REPOSITORIES	TECHNICAL ARTICLES	CORE DISCIPLINES	GLOBAL REACH
22 Active Repos	**3 Published Guides**	**AI, Vision, Next.js, PWA**	**Open-Source Access**
github.com/basseyriman	medium.com/@basseyriman	Production-Grade Engineering	Free Reusable MIT/Public Code

01 | FEATURED TECHNICAL PUBLICATIONS & PEER DEVELOPER GUIDES ON MEDIUM

Article 01: Building a Premium Budget Tracker: 3 Technical Challenges in React

URL: https://medium.com/@basseyriman/building-a-premium-budget-tracker-3-technical-challenges-and-how-i-solved-them-in-react-9eac01779a41

Deep technical analysis of frontend state synchronization in complex reactive dashboards. Focuses on: (1) Scalable currency input rendering without horizontal overflow via dynamic length-based font scaling; (2) Deterministic category hashing that automatically assigns semantically appropriate icons to arbitrary user input strings; (3) Cross-tab store synchronization using Zustand.

Article 02: Building PursePro: A Privacy-First Budget App Outside the App Store

URL: https://medium.com/@basseyriman/building-pursepro-a-privacy-first-budget-app-that-lives-outside-the-app-store-824e3669107d

Comprehensive architectural guide on building privacy-preserving, app-store-independent financial tools using Progressive Web Application (PWA) standards. Details service worker caching strategies, IndexedDB local persistence, offline encryption, and responsive desktop/mobile layout parity using Tailwind CSS.

Article 03: How to Build a Premium AI Portfolio with Next.js 15 & Framer Motion

URL: https://medium.com/@basseyriman/how-to-build-a-premium-ai-portfolio-with-next-js-15-framer-motion-f22ef697bb40

Step-by-step engineering tutorial instructing AI and machine learning engineers on designing high-performance interactive portfolios. Provides modular implementations of staggered entrance motion variants, hardware-accelerated CSS glassmorphism, and structured component data mapping to communicate deep technical capability.

FIGURE 1.1: PRODUCTION ARTIFACTS & CODE ENGINE SHARED IN ARTICLES

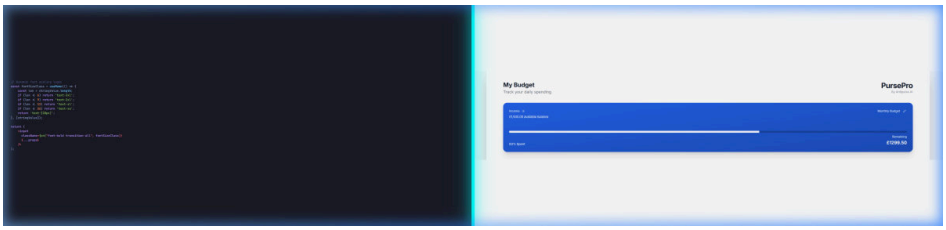


Fig 1.1A: Technical Article Cover & Implementation Artifacts (Dynamic Math Scaling)

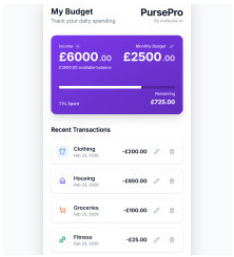


Fig 1.1B: PursePro PWA Dashboard

Community Impact: These articles bridge the gap between academic theory and practical software engineering, providing reusable solutions for developers building modern web apps. The tutorials include live code sandboxes, GitHub references, and interactive demos allowing engineers across the UK and internationally to inspect and adapt the codebase.

OPEN-SOURCE ECOSYSTEM: PUBLIC CODE REPOSITORIES AT GITHUB.COM/BASSEYRIMAN

Bassey Riman maintains 22 public software repositories on GitHub (github.com/basseyriman), actively contributing open-source tools, computer vision models, machine learning benchmarks, and modern web application frameworks to the developer community.
The repositories range from medical deep learning (Vision Transformers for MRI diagnostics) and browser-based AI code intelligence (GitNexus Graph RAG) to interactive educational games (AI Quest) and autonomous browser guidance agents (Pointly).

02 | VERIFIED PUBLIC GITHUB PROFILE & REPOSITORY VELOCITY

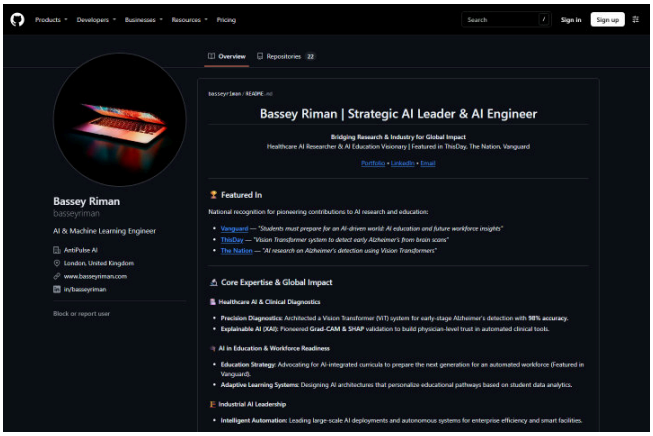


Fig 2.1: GitHub Profile (github.com/basseyriman) featuring National Recognition

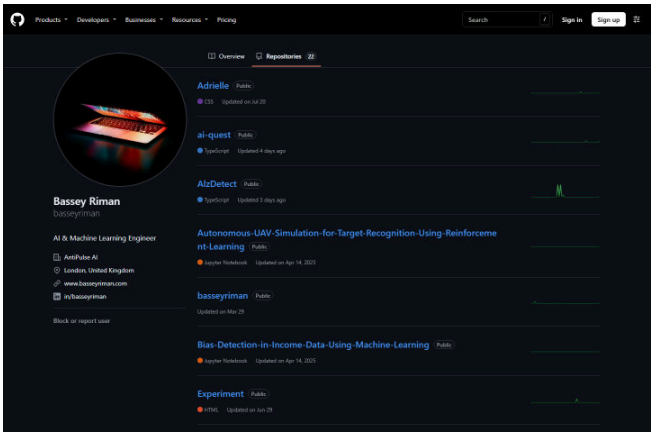


Fig 2.2: Public Repositories (22 Public Repositories across AI & Web)

03 | SELECTION OF PUBLIC OPEN-SOURCE REPOSITORIES & ARCHITECTURAL HIGHLIGHTS

AlzDetect	TypeScript / Python	Explainable ViT-B/32 MRI Alzheimer's detection platform with Grad-CAM and attention rollout.
ai-quest	TypeScript / React	Interactive 7-stage ML simulation game teaching machine learning pipelines to developers.
GitNexus	Client-Side RAG	Zero-server code intelligence knowledge graph creator with built-in client-side Graph RAG Agent.
Pointly	TypeScript / Node	Chrome MV3 AI navigation assistant guiding users through web workflows with zero auto-clicks.
VoiceDress	TypeScript / Next 16	Voice-orchestrated personal styling and virtual try-on engine powered by diffusion models.
Young-AI-Explorers	TypeScript / Next 16	AI-native educational curriculum platform with 38 interactive lessons across 6 learning islands.
PursePro	TypeScript / PWA	Privacy-first progressive web application for personal budget management and cash-flow tracking.
Autonomous-UAV-Simulation	Jupyter / PyTorch	Deep reinforcement learning simulation for autonomous UAV target recognition in complex terrain.

04 | REUSABLE CODE IMPLEMENTATIONS PUBLISHED FOR THE DEVELOPER COMMUNITY

ScalableAmountInput.tsx

```tsx
// Dynamic font scaling logic
const fontSizeClass = useMemo(() => {
    const len = stringValue.length;
    if (len <= 6) return 'text-3xl';
    if (len <= 9) return 'text-2xl';
    if (len <= 12) return 'text-xl';
    if (len <= 26) return 'text-xs';
    return 'text-[10px]';
}, [stringValue]);

return (
    <input
      className={cn("font-bold transition-all", fontSizeClass)}
      {...props}
    />
);
```

Snippet 1: Dynamic Font Scaling Logic (ScalableAmountInput.tsx)

categoryUtils.tsx

```tsx
// Deterministic hashing for fallback icons
const stringToHash = (str: string) => {
    let hash = 0;
    for (let i = 0; i < str.length; i++) {
        const char = str.charCodeAt(i);
        hash = (hash << 5) - hash + char;
        hash = hash & hash;
    }
    return Math.abs(hash);
};

const hash = stringToHash(lowerCat);
const iconIndex = hash % GENERIC_ICONS.length;
const icon = GENERIC_ICONS[iconIndex];
```

Snippet 2: Deterministic Hashing for Dynamic Icons (categoryUtils.tsx)

src/components/skills-section.tsx

```tsx
const containerVariants = {
  hidden: { opacity: 0 },
  visible: {
    opacity: 1,
    transition: {
      staggerChildren: 0.1,
    },
  },
}

const itemVariants = {
  hidden: { opacity: 0, scale: 0.95 },
  visible: {
    opacity: 1,
    scale: 1,
```

Snippet 3: Staggered Entrance Variants (skills-section.tsx)

src/app/globals.css

```css
.glass {
  @apply bg-background/40 backdrop-blur-xl border border-white/10;
  box-shadow: 0 8px 32px 0 rgba(0, 0, 0, 0.37);
}
```

Snippet 4: Reusable Hardware-Accelerated Glass CSS Utility (globals.css)

05 | PUBLIC RECORD, CITATION REGISTRY & COMMUNITY ADVANCEMENT METRICS

Medium Profile & Publication Hub	https://medium.com/@basseyriman	Public technical articles on React, Next.js, and ML architectures
Article 1: React State & UI Scaling	https://medium.com/@basseyriman/building-a-premium-budget...	Published in-depth technical analysis of UI math scaling
Article 2: Privacy-First PWA Systems	https://medium.com/@basseyriman/building-pursepro-a-priva...	Published guide on offline IndexedDB & zero-tracking PWAs
Article 3: Next.js 15 AI Engineering UI	https://medium.com/@basseyriman/how-to-build-a-premium-ai...	Published tutorial on declarative animation & glassmorphism
GitHub Developer Profile	https://github.com/basseyriman	22 public open-source repositories spanning AI, CV & Web
GitNexus Knowledge Graph Engine	https://github.com/basseyriman/GitNexus	Client-side Graph RAG code intelligence engine for developers
STEM Educational Mentorship	UK Primary Schools & STEM Outreach	Author of published AI book & school library donor